Theory case · Topic 2 · 2 points · 15 min

C15 — LowOrbit: the drivers behind a cost-leadership strategy in satellite buses

LowOrbit S.A. sells the "bus" (platform) of small Earth-observation satellites at 40% below its rivals. It builds one standard design with interchangeable modules instead of custom platforms, and has designed every part to be machined in a single setup. Its line now produces 120 buses a year, and the hours per bus have fallen by half since the first 20 units as the teams repeated the same tasks. It bought a solar-cell supplier located next to the plant, keeps the factory running in three shifts, and its managers are paid for removing waste in every process.

Part (a) · 1.25 points

Identify the drivers of cost advantage present in LowOrbit

Driver [Slides 21-22]	Evidence at LowOrbit
Product design	Standardization of designs and components; parts designed for manufacture (single setup)
Economies of scale	120 buses a year: specialization and division of labour
Economies of learning	Hours per bus halved as routines improved
Input costs	Ownership of a low-cost input (the solar-cell supplier) and a location advantage
Capacity utilization	Three shifts spread the fixed costs over more units
Residual efficiency	Managerial efficiency, motivation and culture of eliminating waste

KEY POINTS

- Product design: one standard design, interchangeable modules, design for manufacture.
- Economies of scale: 120 buses a year.
- Economies of learning: hours per bus halved with repetition.
- Input costs: ownership of a low-cost input (solar cells), location.
- Capacity utilization: three shifts.
- Residual efficiency: managers paid to remove waste (motivation, culture).

Part (b) · 0.75 points

Explain what a cost advantage is and what risk LowOrbit runs with this strategy

A **cost advantage** means offering a **similar product at a lower cost** than competitors, whereas a differentiation advantage obtains a **price premium from a unique product** [Slide 20]. LowOrbit's standard bus is its main cost driver, but it is also its main risk: customers whose missions need a custom platform will go to differentiated rivals, and competitors can copy standardization and scale. To keep the advantage it must keep reducing costs faster than its rivals (learning, scale) while keeping the product "similar enough" in performance to theirs.

KEY POINTS

- Cost advantage: offering a similar product at a lower cost than rivals.
- Risk: standardization limits customization; customers who value unique features may prefer differentiated rivals; rivals may copy the cost drivers.